$$\log \frac{H_2PO_4^-}{[HPO_4^{2-}]} = -2.0$$

Converting log term in the above equation

$$\frac{H_2PO_4^-}{[HPO_4^{2-}]} = 10^{-2}$$

or

$$\frac{H_2PO_4^-}{[HPO_4^{2-}]} = \frac{1}{100}$$

Rearranging equation

$$[HPO_4^{2-}] = 100\ [H_2PO_4^-]$$

The concentration of $[H_2PO_4^{2-}]$ is 100 times of concentration of $[HPO_4^-]$ to get pH of 4.8 as per Henderson–Hasselbalch equation.

EVALUATION

I. Part A

1) Which one of the following is a membrane less organelle?

a. nucleus b. nucleolus c. endoplasmic reticulum

d. mitochondria.

2) The nature of the lipid bilayer is

a. hydrophilic b. hydrophobic c. amphipathic

d. lipophilic

3) The cell wall is made up of

a. glucose b. cellulose c. sucrose d. fructose

4) The suicidal bag of the cell is

a. mitochondria b. nucleus c. lysosome

d. endoplasmic reticulum

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